

Midterm Review Packet

Topics 3.1–3.10

**Also study**

Your Unit 1 and Unit 2 Review Packets, Quizzes 1–6, and Pseudocode Bootcamp Levels 0–3.

Tracing & Concepts

1. This code is meant to swap `x` and `y`:

```
x ← y
y ← x
```

What does it actually do?

- A. Swaps `x` and `y` correctly
- B. Both variables end with the original value of `x`
- C. Both variables end with the original value of `y`
- D. Causes an error

2. Which TWO code segments correctly swap the values of `a` and `b`? (*Select two.*)

```
I.  temp ← a
    a ← b
    b ← temp

II. a ← b
    b ← a

III. a ← a + b
     b ← a - b
     a ← a - b

IV.  temp ← a
     a ← b
     b ← a
```

- A. I
- B. II
- C. III
- D. IV

3. Using the AP reference sheet, what does `a` contain after this code?

```
a ← [2, 4, 6, 8]
b ← a
b[2] ← 10
```

- A. [2, 4, 6, 8]
- B. [2, 10, 6, 8]
- C. [10, 4, 6, 8]
- D. [2, 4, 10, 8]

4. Which expression ALWAYS gives the last element of a non-empty list `nums`?

- A. `nums[LENGTH(nums) - 1]`
- B. `nums[LENGTH(nums)]`
- C. `nums[0]`
- D. `nums[LENGTH(nums) + 1]`

5. What is the value of `4 + 8 MOD 3 * 2`?

- A. 6
- B. 8
- C. 12
- D. 2

Midterm Review Packet

continued

Tracing & Concepts

6. What is displayed?

```
a ← 7
b ← 3
a ← a MOD b + a
b ← a MOD (b + 2)
a ← (a + b) MOD 4
DISPLAY(a)
DISPLAY(b)
```

- A. 0 1
- B. 2 3
- C. 3 3
- D. 3 8

7. How many characters are in the string "AP CSP 2027" ?

- A. 11
- B. 10
- C. 9
- D. 8

8. `len(str)` returns the number of characters in `str`. What is displayed?

```
word ← "STAR"
result ← ""
i ← len(word)
REPEAT UNTIL (i < 1)
{
    result ← concat(result,
        substring(word, i, 1))
    i ← i - 1
}
DISPLAY(result)
```

- A. STAR
- B. RATS
- C. SSSS
- D. Error

9. Which expression is equivalent to `NOT (x > 10 OR y < 5)` ?

- A. $(x < 10) \text{ OR } (y > 5)$
- B. $(x < 10) \text{ AND } (y > 5)$
- C. $(x \leq 10) \text{ OR } (y \geq 5)$
- D. $(x \leq 10) \text{ AND } (y \geq 5)$

10. A store gives a discount if a customer is a member AND spent at least \$50, OR if it's the customer's birthday. Which expression is correct?

- A. `isMember AND (spent ≥ 50 OR isBirthday)`
- B. `(isMember AND spent ≥ 50) OR isBirthday`
- C. `isMember OR spent ≥ 50 OR isBirthday`
- D. `(isMember OR spent ≥ 50) AND isBirthday`

Midterm Review Packet

continued

Tracing & Concepts

11. Which segments display "big" when $x > 10$ and "small" otherwise, for ALL integers x ?

- I. IF ($x > 10$) { DISPLAY("big") }
ELSE { DISPLAY("small") }
- II. IF ($x > 10$) { DISPLAY("big") }
IF ($x < 10$) { DISPLAY("small") }
- III. IF ($x \leq 10$) { DISPLAY("small") }
ELSE { DISPLAY("big") }

- A. I only
- B. II only
- C. I and III only
- D. I, II, and III

12. What is displayed?

```
a ← 5
b ← 10
IF (a > b) { a ← b }
ELSE { b ← a }
IF (a = b) { a ← a + b }
DISPLAY(a)
DISPLAY(b)
```

- A. 10 5
- B. 5 10
- C. 10 10
- D. 15 10

13. Which statement is true about this code?

```
cost ← 5
IF (weight > 10)
{ cost ← 20 }
ELSE
{
  IF (isMember) { cost ← 0 }
}
```

- A. A non-member with weight 4 pays 5
- B. A member with weight 12 pays 0
- C. A member with weight 10 pays 20
- D. Every member pays 0

14. For which TWO inputs is "Go" displayed?
(Select two.)

```
IF (NOT (light = "red"))
{
  IF (light = "green" OR clear)
  { DISPLAY("Go") }
}
```

- A. light = "red", clear = true
- B. light = "green", clear = false
- C. light = "yellow", clear = true
- D. light = "yellow", clear = false

Midterm Review Packet

continued

Tracing & Concepts

15. What is displayed?

```
result ← 0
i ← 1
REPEAT UNTIL (i > 10)
{
  IF (i MOD 2 = 0)
  { result ← result + i }
  i ← i + 1
}
DISPLAY(result)
```

- A. 25
- B. 20
- C. 55
- D. 30

16. This code should compute the product of all numbers in `nums`, but always displays 0. Which change fixes it?

```
product ← 0
FOR EACH n IN nums
{ product ← product * n }
DISPLAY(product)
```

- A. Change `product * n` to `product + n`
- B. Move `DISPLAY` inside the loop
- C. Initialize `product` to 1 instead of 0
- D. Change `FOR EACH` to `REPEAT UNTIL`

17. A robot faces **up**. Where does it end up?

```
REPEAT 3 TIMES
{
  MOVE_FORWARD()
  ROTATE_RIGHT()
}
```

- A. Back at the start, facing up
- B. One square right of the start, facing left
- C. One square above the start, facing right
- D. Three squares above the start, facing up

18. Which TWO segments set `max` to the larger of `a` and `b` for ALL values, including when `a = b`? (*Select two.*)

```
I. IF (a > b) { max ← a }
   ELSE { max ← b }
II. max ← a
   IF (b > max) { max ← b }
III. IF (a > b) { max ← a }
     IF (b > a) { max ← b }
IV. max ← b
    IF (a < b) { max ← a }
```

- A. I
- B. II
- C. III
- D. IV

Name: _____

Class: _____

Date: _____

Midterm Review Packet

continued

Tracing & Concepts

19. This code is meant to find the minimum of nums.

```
min ← 0
FOR EACH n IN nums
{
  IF (n < min) { min ← n }
}
```

For which list does it work correctly?

- A. [5, 3, 9]
- B. [-2, -8, 4]
- C. [1, 2, 3]
- D. [7]

20. What is result after this code?

```
list1 ← [10, 20, 30, 40]
result ← []
FOR EACH x IN list1
{
  IF (x MOD 20 = 0)
  { APPEND(result, x / 10) }
  ELSE
  { APPEND(result, x) }
}
```

- A. [1, 2, 3, 4]
- B. [2, 4]
- C. [10, 2, 30, 4]
- D. [10, 20, 30, 40]